

Suricata Dashboard

📅 Date	July 28, 2026
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Suricata Dashboard — Session Notes

What changed from yesterday

Yesterday's version was a working but basic dashboard: raw eve.json read directly into memory (`session_state`), simple multiselect filters, alerts and traffic living in the same rolling in-memory window with no persistence. Today's work moved the dashboard from "working prototype" to "persistent, feature-rich tool" — the biggest shift being that **alerts now live in a real SQLite database instead of only in memory**, which unlocked review workflow, sorting, and survival across browser refreshes.

New features added today

1. Persistent alert storage (SQLite)

- Alerts are now written to `alerts.db` instead of only being held in `session_state`.
- This means alerts survive a browser refresh, a Streamlit restart, or even a VM reboot — previously, refreshing the page reset everything to zero.

2. Alert review workflow

- Added "Mark Reviewed" and "Mark False Positive" buttons on each alert.
- Reviewed/FP alerts drop out of the default view automatically.
- A sidebar toggle ("Show reviewed / false-positive alerts") brings them back, with a "Reopen" button to undo.

3. Sorting by urgency

- Learned how Suricata's severity actually works: **severity 1 = most severe/highest priority, 3 = lowest** — opposite of what you'd naturally

assume ("bigger number = worse" is wrong here).

- Added a "Sort by" dropdown: Most urgent first (severity ascending), Newest first, Oldest first, Confidence (High → Low).

4. Grouped alert view

- Added a toggle to collapse repeated alerts (same signature + source IP) into a single summary row showing count, first seen, last seen — useful when one noisy signature fires dozens of times.

5. Regex search support

- Signature/IP text filters now have an optional "Regex" checkbox next to them, instead of only plain substring matching.
- Falls back safely (with a warning) if the typed regex pattern is invalid, instead of crashing.

6. Expanded filter set

- Added filters that didn't exist before: Signature Severity, Confidence, Attack Target, Interface, Direction, Flow ID, byte-count ranges (bytes to server/client), Signature ID search.
- Filters were also grouped into collapsible sidebar sections (Severity/Confidence/Target, Protocol/Category/Signature, IPs/Ports, Interface/Flow/Direction) instead of one long flat list.

7. CSS/visual polish

- Added colored severity "chips" (red/amber/blue badges) instead of plain text labels.
- Styled metric cards and expanders with subtle borders/rounded corners for a more consistent, less default-Streamlit look.

8. CSV export

- Added "Download filtered results as CSV" buttons on both the Alerts and Traffic tabs, respecting whatever filters are currently applied.

Big bugs found and fixed today

1. Two files silently fighting over the same database schema

- `ingestor.py` (the file actually running) was creating the `alerts` table with only 8 basic columns.
- `alert_store.py` defined a much richer schema for the *same* table name — but since SQLite's `CREATE TABLE IF NOT EXISTS` does nothing if the table already exists, the thin schema from `ingestor.py` silently won.
- Result: filters like Category, Signature Severity, Confidence, Attack Target, Interface, Direction all appeared empty — not because of missing data, but because those columns never existed in the actual running database.
- **Fix:** consolidated to one single source of truth — `ingestor.py` now owns the full schema (alerts + traffic + file-offset tracking), and `alert_store.py` was stripped down to just the one thing it's actually used for (updating an alert's review status).
- **Required a one-time database reset** (`rm -f alerts.db alerts.db-wal alerts.db-shm`) since the old thin table wouldn't automatically gain the new columns.

2. Traffic table flickering on every refresh

- Root cause: Streamlit's `@st.fragment(run_every=...)` re-executes the *entire* function body on every tick, and any `st.dataframe()` call gets fully torn down and rebuilt — that's what caused the visible flash every few seconds.
- Tried a "fingerprint" approach (skip redrawing if nothing changed) but on closer inspection this didn't actually gate the render — it only tracked a variable.
- **Landed on:** giving the dataframe a stable `key="traffic_table"`, which helps Streamlit reuse the existing widget rather than fully recreating it. Full elimination of flicker on *genuine* new data would need a heavier component like `streamlit-aggrid` — flagged as a future improvement, not done today.

3. Dashboard froze on a gray "loading skeleton" screen

- Turned out to be a normal Streamlit/browser hiccup (WebSocket disconnect or a stuck fragment render), not an internet problem — worth noting since it looked network-related but wasn't (the dashboard is accessed over the local VM network, not the actual internet).

- Resolved by refreshing the browser tab / restarting Streamlit.

4. Sidebar showing both tabs' filters at once ("filter bleed")

- Discovered that in Streamlit, code inside *both* `with tab_alerts:` and `with tab_traffic:` blocks runs on every single script execution regardless of which tab is visually active — only the displayed *content area* is tab-specific, not the code execution.
 - Since both `sidebar_alert_filters()` and `sidebar_traffic_filters()` write into `st.sidebar`, both filter sections always appear stacked together, which is why Alert Filters showed up even while viewing the Traffic tab.
 - **Identified, not yet fixed** — next step is to only render the relevant tab's filters based on which tab is active.
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Key concepts learned/clarified today

- **Suricata severity direction:** 1 = high/most severe, 3 = low — not the intuitive direction.
 - **Where "tags" like Signature Severity/Confidence actually come from:** not computed by the dashboard at all — they're written directly into the rule file by the rule author (e.g. Proofpoint/ET) and Suricata just echoes them into `eve.json`'s `alert.metadata` block.
 - **How to independently verify filter accuracy:** cross-check a filtered count in the dashboard against a direct SQL query (`sqlite3 alerts.db "SELECT COUNT(*) FROM alerts WHERE ..."`) rather than trusting the UI blindly.
 - **Local network vs internet access:** accessing the dashboard via `192.168.23.159:8501` is local VM-network traffic, not internet-dependent — useful to remember when diagnosing "is it slow/broken because of internet" type questions.
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Still open / next steps

- Fix the sidebar filter-bleed issue (Alert filters showing on Traffic tab and vice versa).

- Decide whether to pursue `streamlit-aggrid` for true flicker-free live table updates.
- Continue the ET rule scraping (Proofpoint threat intel) in the background — separate task, unaffected by today's dashboard changes.